yolo_ros

Provides a ROS package based on PyTorch-YOLO.

Running Environment:

- Ubuntu 20.04
- ROS Noetic
- Python >= 3.7.0, PyTorch >= 1.7

Environment Configuration:

1. First, install the environment with the corresponding Python and PyTorch versions.
2. Then install the following dependencies:

   ```
   pip install ultralytics
   pip install rospkg
   ```

3. Install Yolo_ROS:

   ```
   cd catkin_ws/src
   git clone https://github.com/HT-hlf/yolo_ros.git
   cd ..
   catkin_make
   ```

Usage

Use YOLO to Detect Objects in Images

Use in Simulation Environment

- Start the simulation environment and robot:

  ```
  source devel/setup.bash
  roslaunch turtlebot3_gazebo turtlebot3_empty_world.launch
  ```

- Start the ros_yolo node:

  ```
  source devel/setup.bash
  roslaunch yolo_ros yolo.launch
  ```

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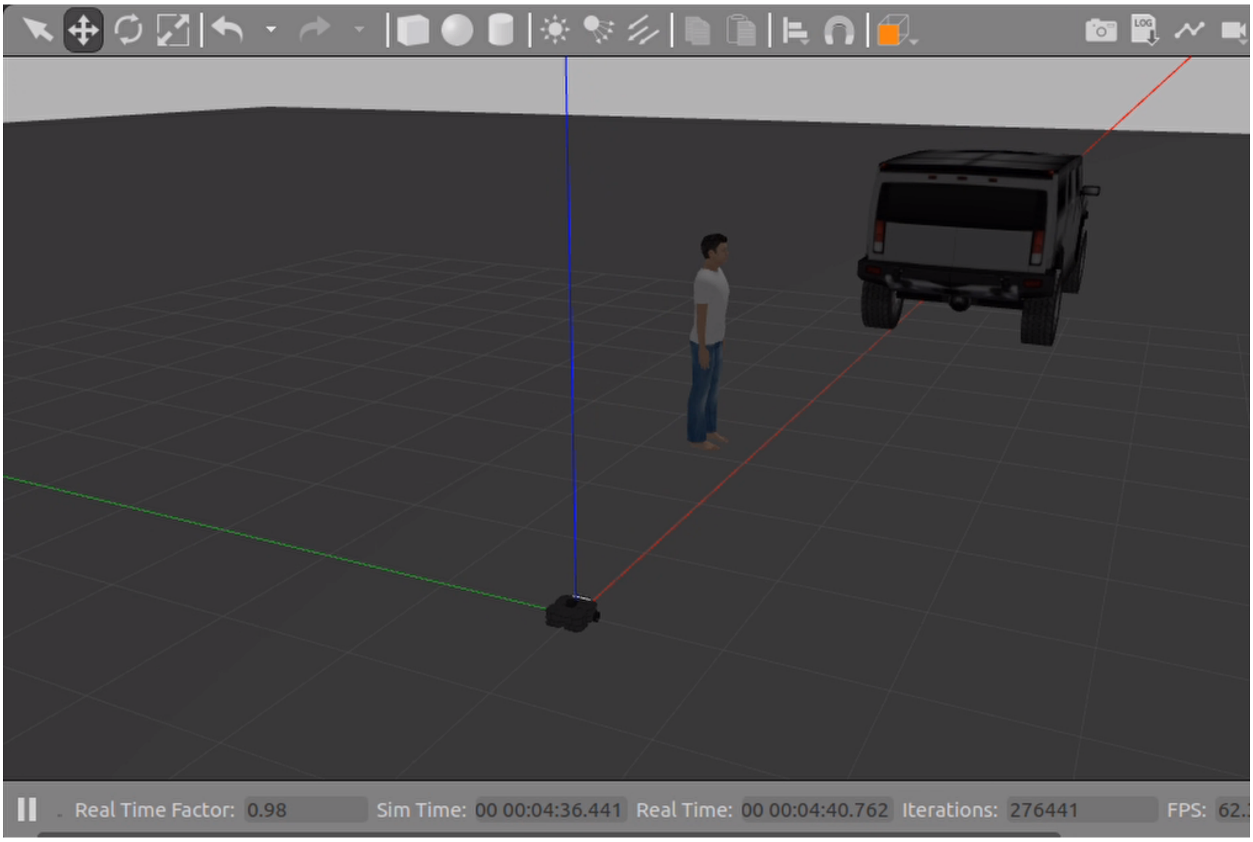
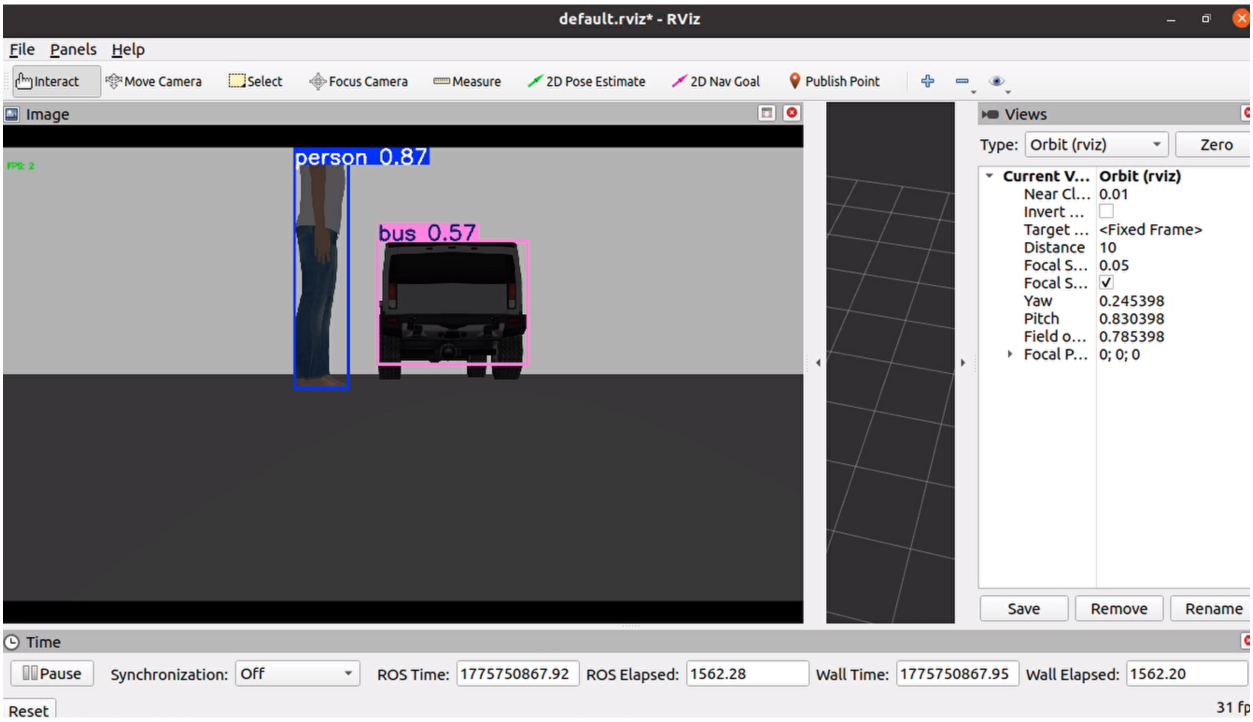




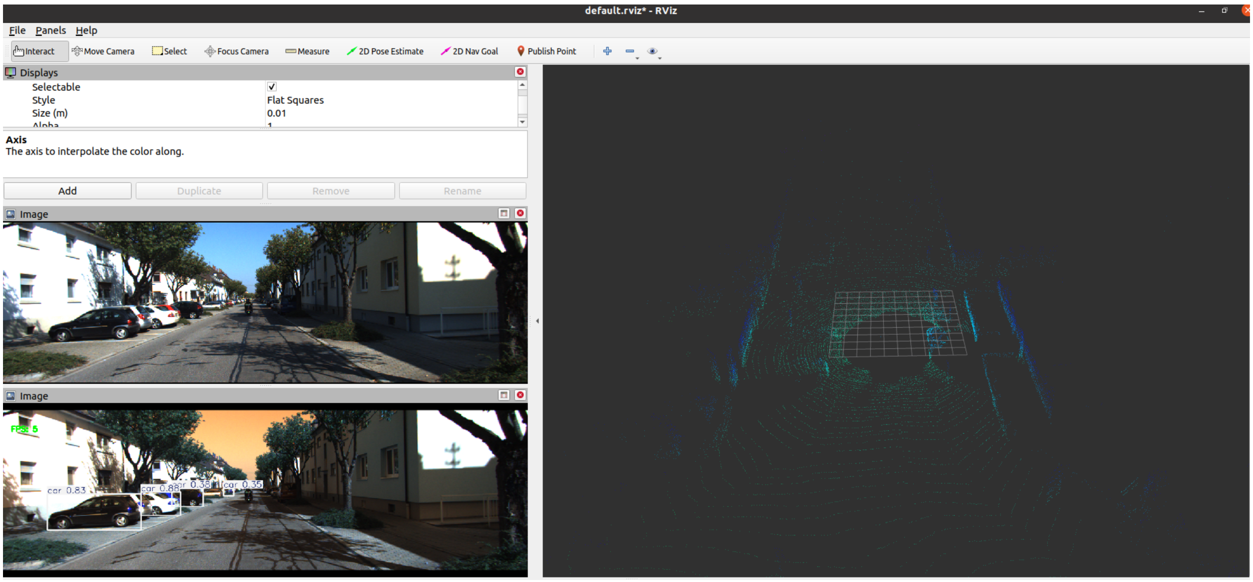
Use with Recorded Dataset

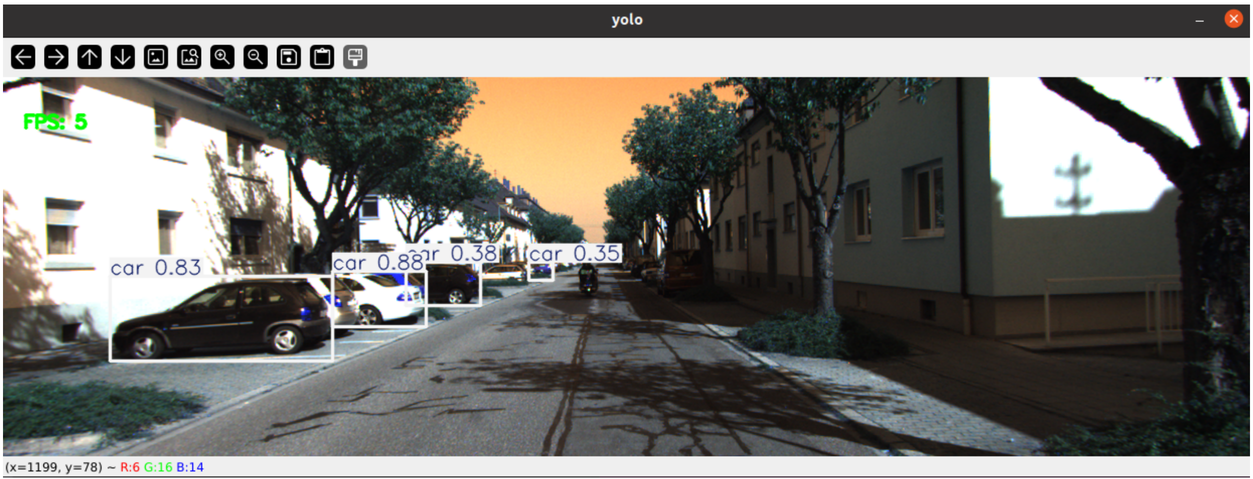
- Play bag data:

```
rosbag play kitti.bag -r 0.2
```

- Start the ros_yolo node:

```
source devel/setup.bash
roslaunch yolo_ros yolo_kitti.launch
```





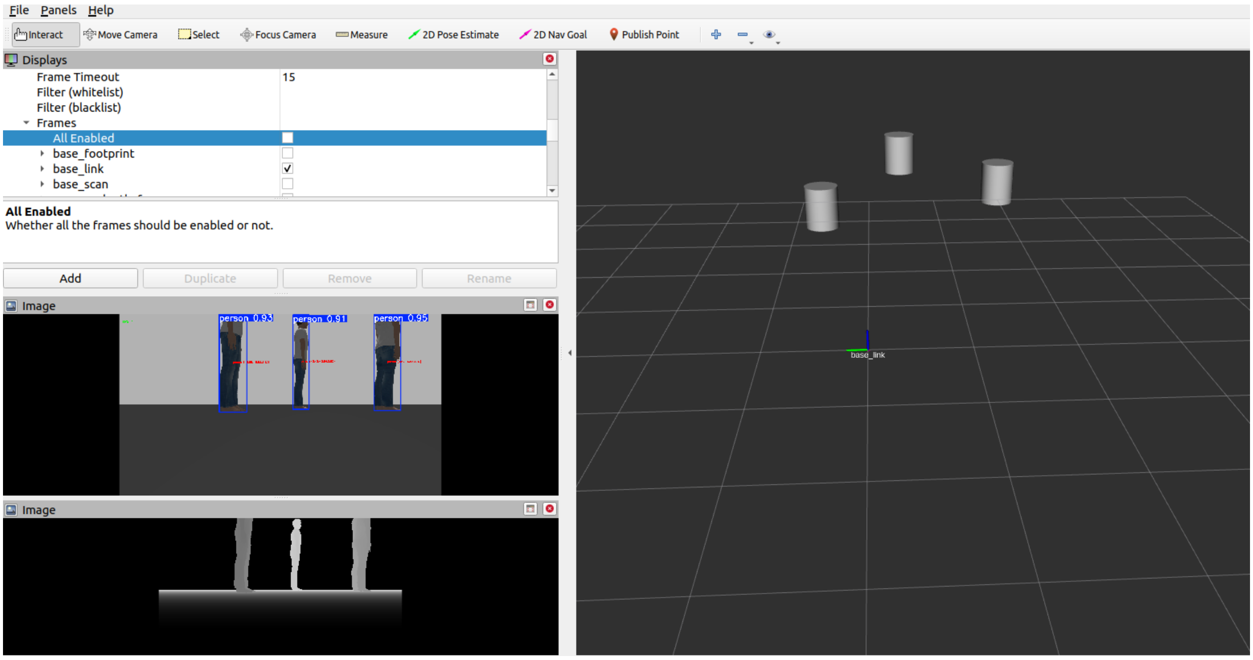
Use YOLO to Detect Objects in Images and Calculate Their 3D Coordinates

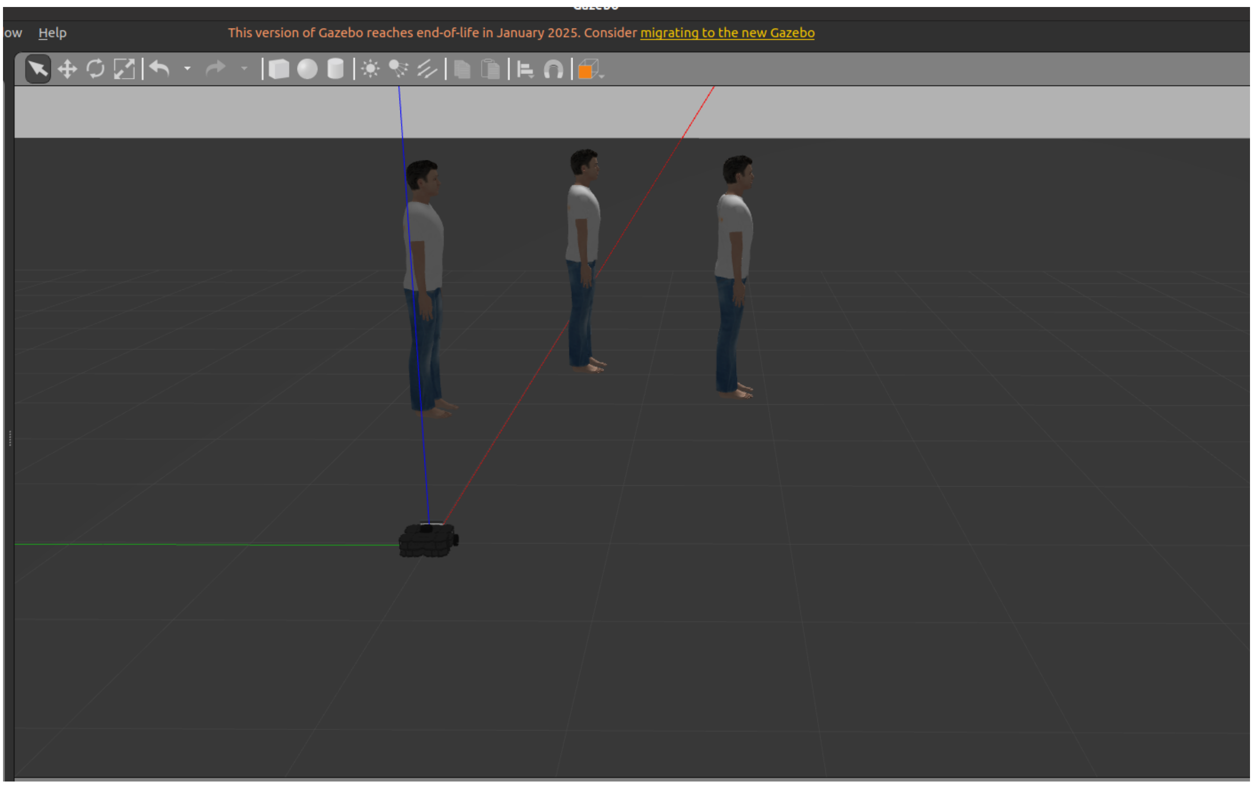
- Start the simulation environment and robot:

```
source devel/setup.bash
roslaunch turtlebot3_gazebo turtlebot3_empty_world.launch
```

- Start the ros_yolo node:

```
source devel/setup.bash
roslaunch yolo_ros yolo_3D.launch
```


File Panels Help
Interact
Move Camera
Select
Focus Camera
Measure
2D Pose Estimate
2D Nav Goal
Publish Point
Displays
Frame Timeout
15
Filter (whitelist)
Filter (blacklist)
Frames
All Enabled
base_footprint
base_link
base_scan
All Enabled
Whether all the frames should be enabled or not.
Add
Duplicate
Remove
Rename
Image
person 0.93
person 0.91
person 0.95
Image
base_link


ow Help
This version of Gazebo reaches end-of-life in January 2025. Consider migrating to the new Gazebo